

Farthest-Point Hypothesis Pruning for Fast Frame-0 Registration in FoundationPose

Phạm Đức Dương, Hoàng Minh Khoa, Nguyễn Đình Cao Minh, Nguyễn Gia Bình,
Trần Hữu Lộc, Biện Minh Trí, Chan Dai Truyen Thai

*Department of Electrical and Computer Engineering
Vietnamese-German University (VGU), Vietnam*

Abstract

FoundationPose registers a novel object on the first video frame by refining and scoring a fixed grid of 252 pose hypotheses; this step dominates initialization latency. This study focuses on how few hypotheses are needed and how they should be selected. From a fixed pool of 504 rendered templates, this paper compares three selection strategies which are uniform random, geodesic farthest-point sampling (FPS), and DINOv2 bag-of-words retrieval. These strategies feed the same FoundationPose refiner and scorer, and are evaluated on 21 YCB-Video objects with ADD/ADD-S against ground truth. Against the full 252-hypothesis grid, FPS with $K=20$ reaches 97.5% ADD-S against the grid's 98.8% at about nine times lower latency, and matches it exactly (98.8%) at $K=50$; across all 21 objects it attains 99.4% at $K=20$. In contrast, DINOv2 bag-of-words retrieval underperforms even random selection for $K \geq 5$, because bag-of-words ranking is not viewpoint-discriminative and concentrates hypotheses by appearance rather than covering the rotation space. Registration time follows $t \approx 192 + 17.0K$ ms. We provide operating recommendations and a failure analysis on small and symmetric objects.

Keywords: 6D object pose estimation, FoundationPose, hypothesis pruning, farthest-point sampling, DINOv2, render-and-compare, YCB-Video.